

# **Problem Solving Using Python (CSE1012)**

Dr. Abdul Kayom, Assistant Professor, SENSE

# Agenda / Contents

**01 Operators & Expressions**

**02 Arithmetic Operators**

**03 Comparison (Relational) Operator**

**04 Assignment Operator**

**05 Examples**

# Operators & Expressions

- Operators and expressions are constructs that are used to manipulate the value of operands
- Some basic operators include: +, -, \*, and /
- Ex.  
sum= a+ b,    here, 'a' and 'b' are operands, '+' is the operator

# Operators & Expressions

- Python supports different types of operators which are as follows:
  - Arithmetic operators
  - Comparison (relational) operators
  - Assignment operators
  - Logical operators
  - Unary operators
  - Bitwise operators
  - Membership operators
  - Identity operators

# Arithmetic Operators

- Some basic arithmetic operators are +, -, \*, /, and //
- These operators can be applied on numbers or variables
- Ex. Take a=100 and b=200 in following table

# Arithmetic Operators

| Operator | Description                                                                                                                                                                                                                            | Example                                                                                                                                                                       | Output                 |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| +        | Addition: Adds the operands                                                                                                                                                                                                            | <code>&gt;&gt;&gt; print(a + b)</code>                                                                                                                                        | 300                    |
| -        | Subtraction: Subtracts operand on the right from the operand on the left of the operator                                                                                                                                               | <code>&gt;&gt;&gt; print(a - b)</code>                                                                                                                                        | -100                   |
| *        | Multiplication: Multiplies the operands                                                                                                                                                                                                | <code>&gt;&gt;&gt; print(a * b)</code>                                                                                                                                        | 20000                  |
| /        | Division: Divides operand on the left side of the operator with the operand on its right. The division operator returns the quotient.                                                                                                  | <code>&gt;&gt;&gt; print(b / a)</code>                                                                                                                                        | 2.0                    |
| %        | Modulus: Divides operand on the left side of the operator with the operand on its right. The modulus operator returns the remainder.                                                                                                   | <code>&gt;&gt;&gt; print(b % a)</code>                                                                                                                                        | 0                      |
| //       | Floor Division: Divides the operands and returns the quotient. It also removes the digits after the decimal point. If one of the operands is negative, the result is floored (i.e., rounded away from zero towards negative infinity). | <code>&gt;&gt;&gt; print(12//5)</code><br><code>&gt;&gt;&gt; print( 12.0//5.0)</code><br><code>&gt;&gt;&gt; print(-19//5)</code><br><code>&gt;&gt;&gt; print(-20.0//3)</code> | 2<br>2.0<br>-4<br>-7.0 |
| **       | Exponent: Performs exponential calculation, that is, raises operand on the right side to the operand on the left of the operator.                                                                                                      | <code>&gt;&gt;&gt; print(a**b)</code>                                                                                                                                         | 100 <sup>200</sup>     |

# Comparison (Relational) Operators

- These are used to compare the values on its either sides and determines the relation between them
- Ex. Assuming  $a = 100$  and  $b = 200$ , we can have following comparisons

# Comparison (Relational) Operators

| Operator           | Description                                                                                                                               | Example                                    | Output |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------|
| <code>==</code>    | Returns True if the two values are exactly equal.                                                                                         | <code>&gt;&gt;&gt; print(a == b)</code>    | False  |
| <code>!=</code>    | Returns True if the two values are not equal.                                                                                             | <code>&gt;&gt;&gt; print(a != b)</code>    | True   |
| <code>&gt;</code>  | Returns True if the value at the operand on the left side of the operator is greater than the value on its right side.                    | <code>&gt;&gt;&gt; print(a &gt; b)</code>  | False  |
| <code>&lt;</code>  | Returns True if the value at the operand on the right side of the operator is greater than the value on its left side.                    | <code>&gt;&gt;&gt; print(a &lt; b)</code>  | True   |
| <code>&gt;=</code> | Returns True if the value at the operand on the left side of the operator is either greater than or equal to the value on its right side. | <code>&gt;&gt;&gt; print(a &gt;= b)</code> | False  |
| <code>&lt;=</code> | Returns True if the value at the operand on the right side of the operator is either greater than or equal to the value on its left side. | <code>&gt;&gt;&gt; print(a &lt;= b)</code> | True   |



# Assignment Operators

- Assigns value to the operand
- Shortcut operators: +=, -=, \*=, /=, %=, //=, and \*\*=

# Assignment Operators

**Table 3.5** Assignment and in-place Operator

| Operator | Description                                                                                                                                                                                                                                                                                         | Example                                      |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| =        | Assign value of the operand on the right side of the operator to the operand on the left.                                                                                                                                                                                                           | c = a , assigns value of a to the variable c |
| +=       | Add and assign: Adds the operands on the left and right side of the operator and assigns the result to the operand on the left.                                                                                                                                                                     | a += b is same as a = a + b                  |
| -=       | Subtract and assign: Subtracts operand on the right from the operand on the left of the operator and assigns the result to the operand on the left.                                                                                                                                                 | a -= b is same as a = a - b                  |
| *=       | Multiply and assign: Multiplies the operands and assigns result to the operand on the left side of the operator.                                                                                                                                                                                    | a *= b is same as a = a * b                  |
| /=       | Divide and assign: Divides operand on the left side of the operator with the operand on its right. The division operator returns the quotient. This result is assigned to the operand to the left of the division operator.                                                                         | a /= b is same as a = a / b                  |
| %=       | Modulus and assign: Divides operand on the left side of the operator with the operand on its right. The modulus operator returns the remainder which is then assigned to the operand on the left of the operator.                                                                                   | a %= b is same as a = a % b                  |
| //=      | Floor division: Divides the operands and returns the quotient. It also removes the digits after the decimal point. If one of the operands is negative, the result is floored (rounded away from zero towards negative infinity): the result is assigned to the operand on the left of the operator. | a //= b is same as a = a // b                |
| **=      | Exponent and assign: Performs exponential calculation, that is, raises operand on the right side to the operand on the left of the operator and assigns the result in the left operand.                                                                                                             | a**= b is same as a = a** b                  |

# Assignment Operators

**Example 3.10** Commands to show the application of the += operator on strings

```
>>> str1 ="Good "  
>>> str2 = "Morning"  
>>> str1 += str2  
>>> print(str1)
```

Good Morning

# Thank You

**Dr. Abdul Kayom**



**Asst. Prof, (SENSE)**



**kayomabdul@vitap.ac.in**



**8474860025**